

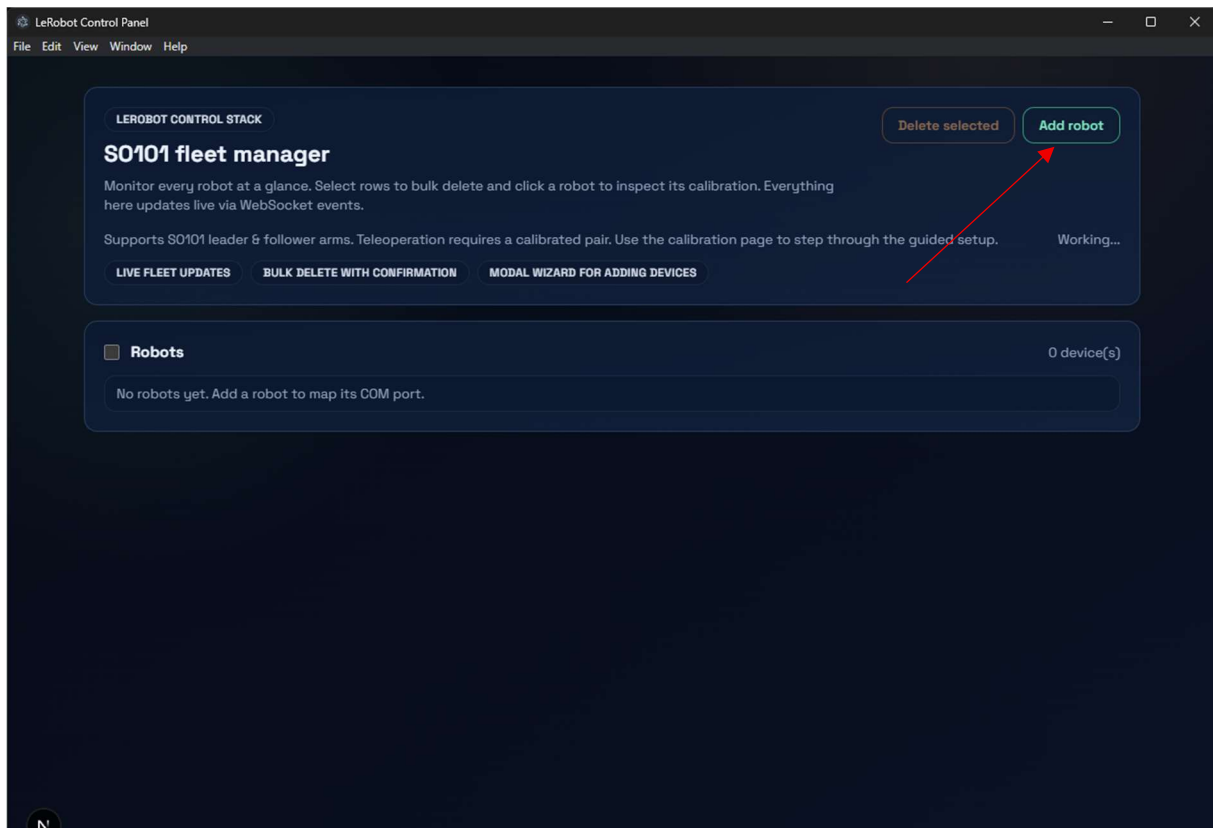
LeRobot Panel Manual

This manual will guide you through every step to achieve the control of your robot arms.

Step 1: Add robot arm

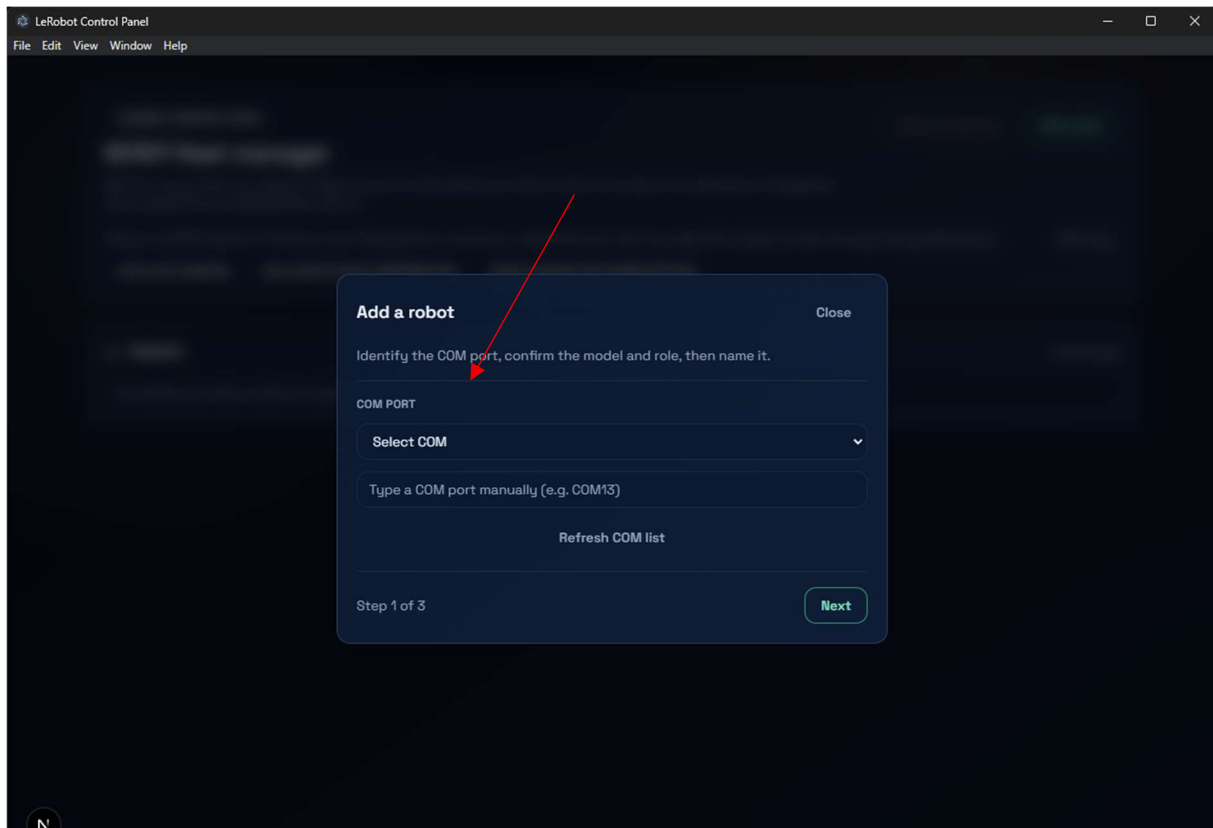
Step 1.1: Click on the “Add robot” button

The first step is to add your robot arm. You can do so by clicking on the add robot button at the top left, on the dashboard page.



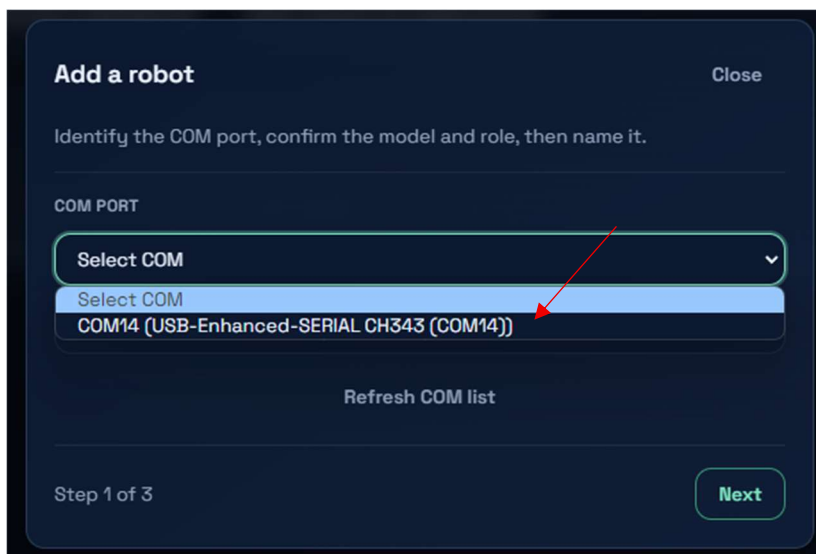
Step 1.2: Select your robot

Great! As you can see, a dialog box opened. In there you can see a “Select COM” dropdown. Click on this item.

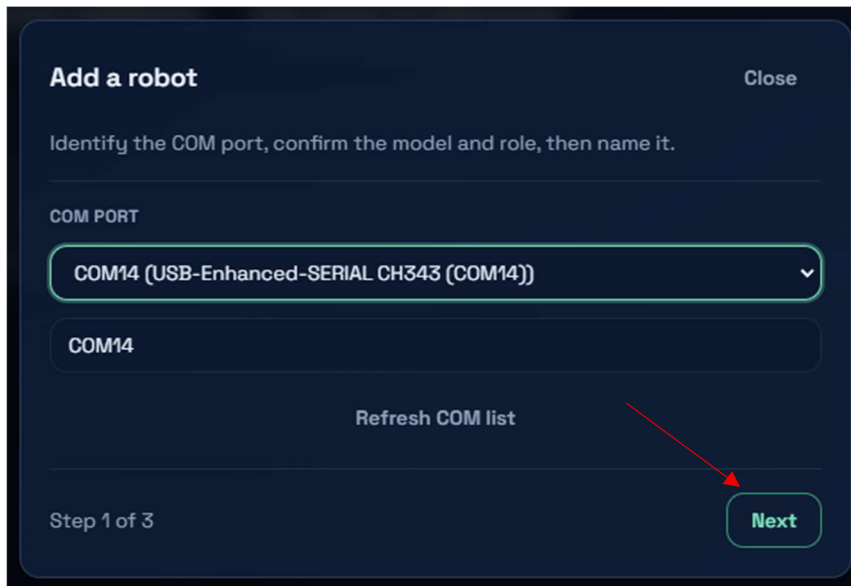


As you can see this will deploy the dropdown. You then should see a list of detected robot arms. Select one of the available options.

NOTE: Do not forget to power the robot arm and to plug it into the usb port on the device running the panel. Otherwise, the robot arm might not be detected.



Once clicked this will select the item and you can proceed by clicking on the “Next” button



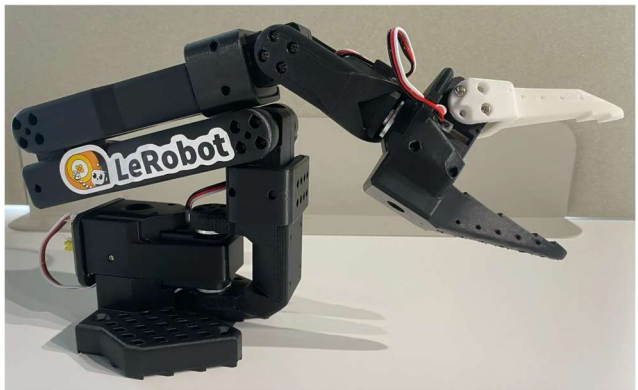
Step 1.3: Select model type and role.

Here you will need to select the model of the device. But since we the program only allows the so101 model for now this part is not needed.

But selecting the role is important. Depending on how the device looks, the role is determined. Here are examples of the two roles.



Leader



Follower

In my case the robot arm is a leader, so I'll be selecting this one. As you can see the selected option is highlighted in green. Once done click on the "Next" button

Add a robot Close

Identify the COM port, confirm the model and role, then name it.

MODEL

S0101

ROLE

Leader arm Follower arm

Step 2 of 3

Back Next

Step 1.4: Define your custom name

Here you will find a input field, select it and give it a custom name. It can be anything you wish, maybe an easy way to recognize it, for example a color.

Add a robot Close

Identify the COM port, confirm the model and role, then name it.

NAME

Friendly name (e.g. Lab Leader)

This name is also used for the calibration file in your lerobot cache.

Step 3 of 3

Back Finish

I'll just call it "Leader", once this is done, click on "Finish" to save your robot arm.

Add a robot Close

Identify the COM port, confirm the model and role, then name it.

NAME

Leader

This name is also used for the calibration file in your lerobot cache.

Step 3 of 3

Back Finish

Step 2: Robot arm calibration

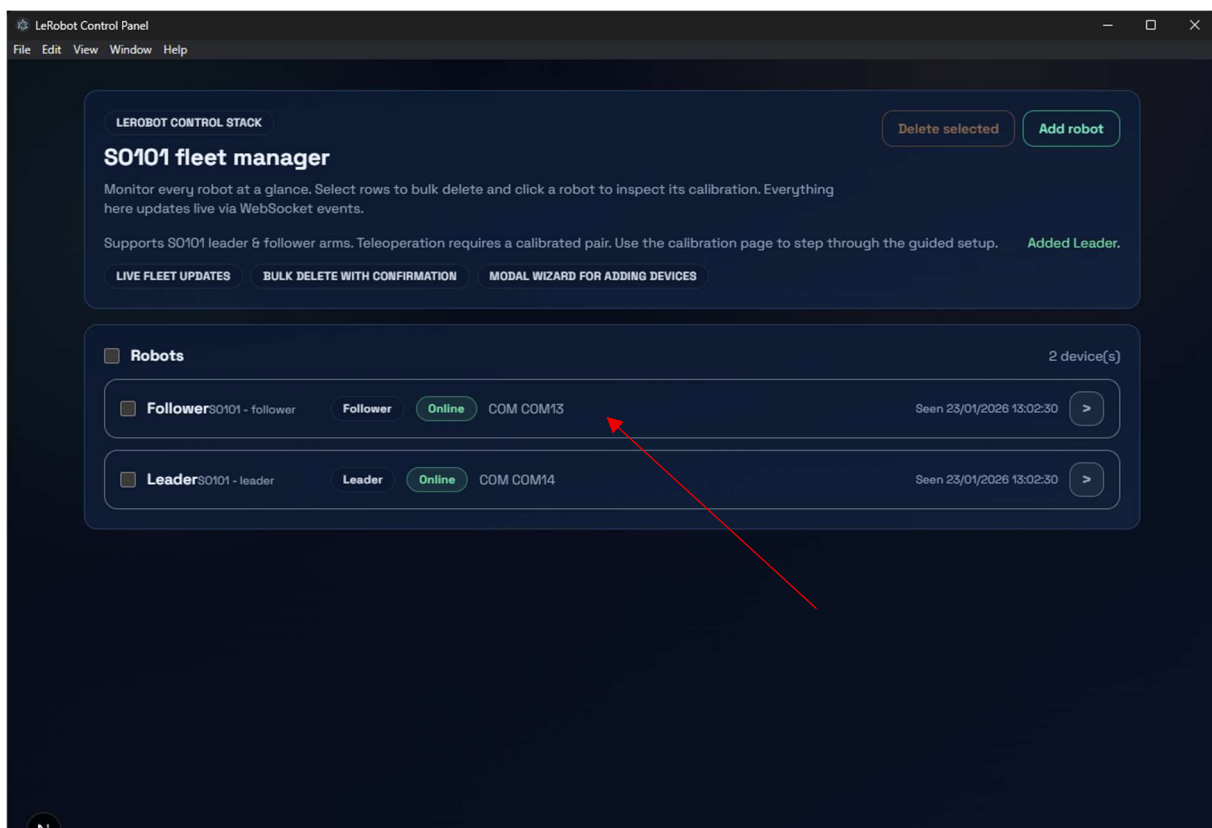
This section is for information only, you can skip to Section 2.1 if you want to start right away.

The calibration step is required. It may sound a bit scary or complicated, but it is actually a very simple concept. Calibration simply means moving all joints of the robot arm to their minimum and maximum positions, that's all.

The software uses this process to learn the movement range of the robot. By recording the minimum and maximum values, it ensures the motors are not forced beyond their safe limits.

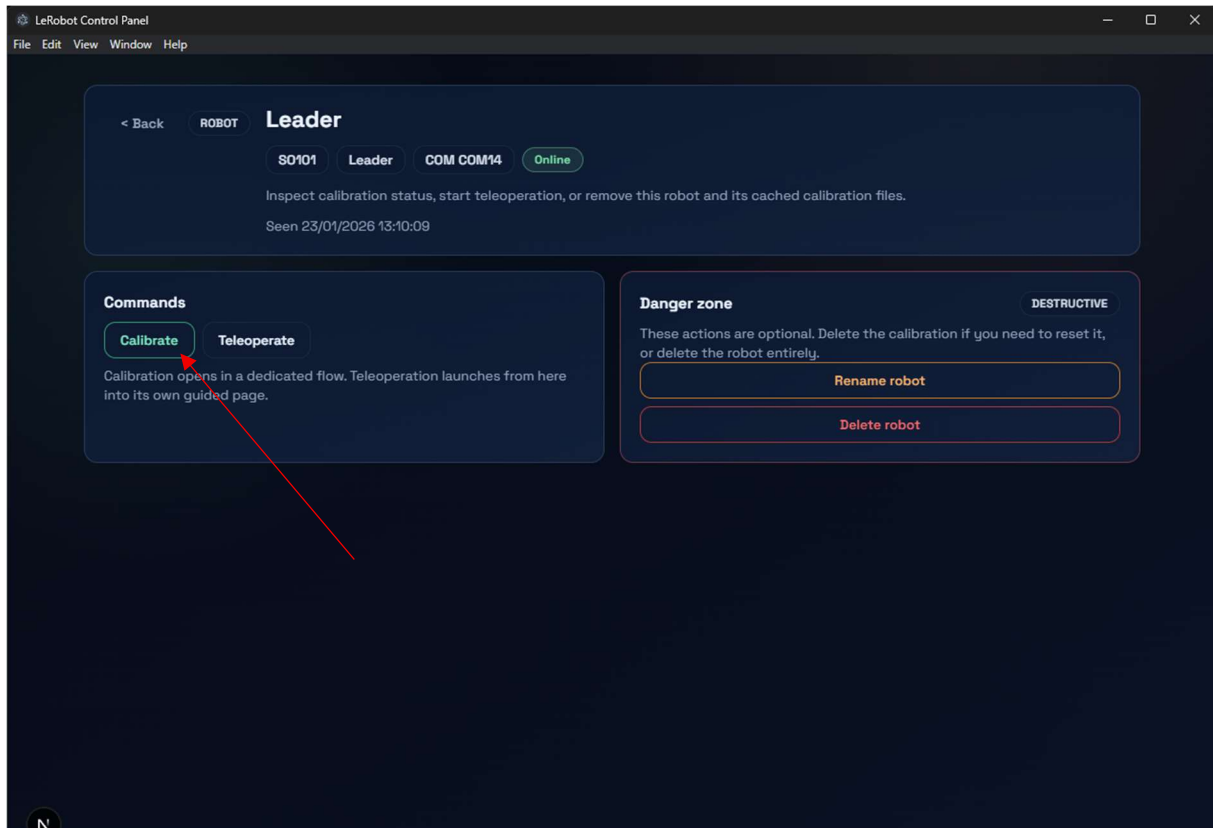
Step 2.1: Access the specific robot page

Don't panic if your panel doesn't look the same, I already added the Follower arm too. We need to access the specific page of the robot we wish to calibrate. Click on the item of the robot you want to calibrate.



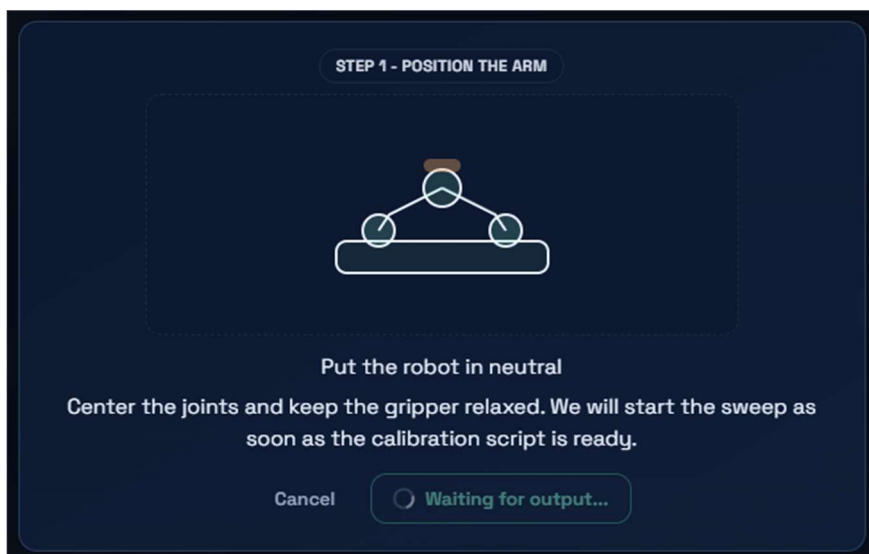
Step 2.2: Access calibration process

To access the calibration process you can just click on the “Calibrate” button.



Step 2.3: Place the robot in neutral position

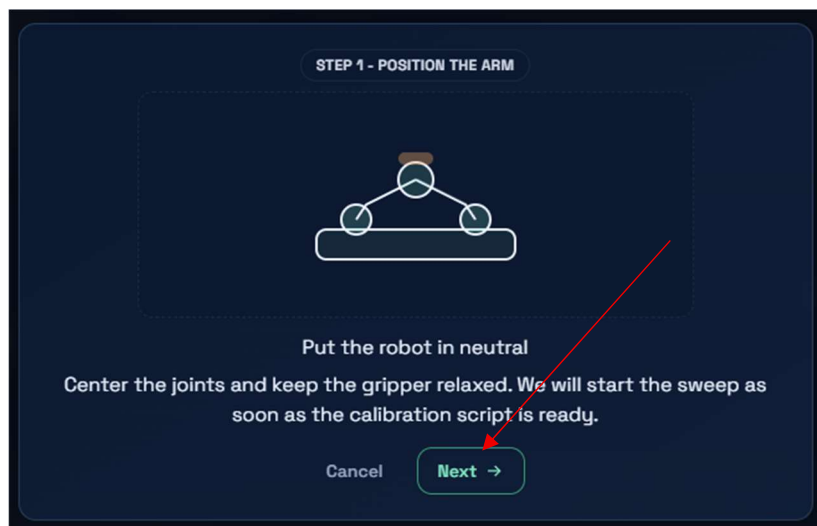
If you see this, you accessed the right place. Don't panic if you see a waiting for output loading button. The software is getting ready, this might take a few seconds, but you can proceed with the next part of this section while this is happening.



Here you are asked to put the robot in neutral position. The neutral position looks like this:



Once you place the robot in that position, hold it and press the “Next ->” button.



Step 2.4: Calibrate the robot

Good, now you need to move every joint to the min and max range. Here is an [example](#). It's easier to do all of these joint by joint.

First, fully open and close the gripper. Then rotate it left and right as far as it can go. Repeat this process for every joint.

As you do this, you will see the slider ranges move from 0 to 100. You must first move each joint to its minimum and maximum positions; this defines the 0 and 100 values.

Once you have gone through all the joints, you can visually check the sliders. They should correctly display the movement range for each joint.

CALIBRATION

Calibrate your robot

< Back

We launch the calibration session right away. Line up the arm during the intro step, then end the sweep once you have moved every joint.

Show output

Calibration started. Move each joint, then hit End calibration to finish.

TARGET ROBOT

Leader

COM COM14 | leader | online

Refresh status

Cancel

Calibration steps

IN PROGRESS

1

Set neutral pose

Place the arm in its neutral position before starting.

NOW

2

Sweep joints

Move each joint through its range and record min/current/max.

3

Save calibration

Confirm values and save to store the calibration.

Live ranges

6 motors

Min/pos/max streamed from the calibration loop. Move joints by hand to update.

MOTOR	MIN	CURRENT	MAX	RANGE / POSITION
shoulder_pan	2047	2047	2051	range 4 0%
shoulder_lift	2036	2047	2047	range 11 100%
elbow_flex	2037	2047	2047	range 10 100%
wrist_flex	2047	2054	2054	range 7 100%
wrist_roll	2043	2046	2047	range 4 75%
gripper	2047	2047	2047	range 0 0%

End calibration

Cancel

End sends a second ENTER to stop the sweep and capture ranges.

Step 2.5: End the calibration

When you're ready to proceed, you can click on the "End calibration" button.

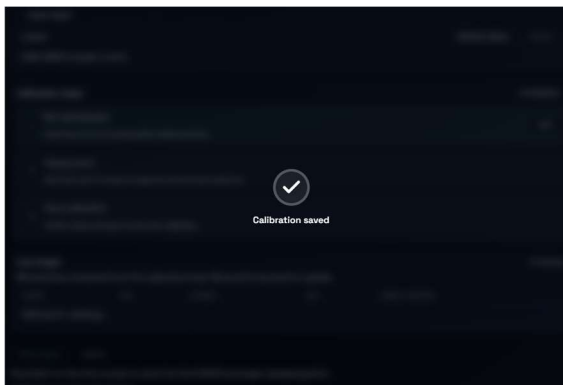
The screenshot shows a calibration interface with two steps: '2 Sweep joints' and '3 Save calibration'. Below these is a 'Live ranges' section for 6 motors. It includes a table with columns: MOTOR, MIN, CURRENT, MAX, and RANGE / POSITION. A red arrow points to the 'End calibration' button at the bottom left.

MOTOR	MIN	CURRENT	MAX	RANGE / POSITION
shoulder_pan	2047	2047	2051	range 4 0%
shoulder_lift	2036	2047	2047	range 11 100%
elbow_flex	2037	2047	2047	range 10 100%
wrist_flex	2047	2054	2054	range 7 100%
wrist_roll	2043	2046	2047	range 4 75%
gripper	2047	2047	2047	range 0 0%

Buttons: End calibration, Cancel

End sends a second ENTER to stop the sweep and capture ranges.

If everything went correctly, you should see this.



NOTE: If you get the page above, but got something like the following image, this means something went wrong, and in my case I need to pay attention to the gripper. Click on "Restart calibration" and go through the calibration process again. Go to step 2.3.

The screenshot shows a "Calibration failed" dialog box. It contains the text: "The calibration process exited with an error. These joints need attention: gripper." The word "gripper" is highlighted with a red box. Below this, it says: "Make sure each joint starts near its mid-range, then move it fully to both extremes before ending calibration." At the bottom are two buttons: "Restart calibration" and "Cancel".

Step 3: Teleoperation

Teleoperation also sounds like a scary word, but this simply means controlling the robot.

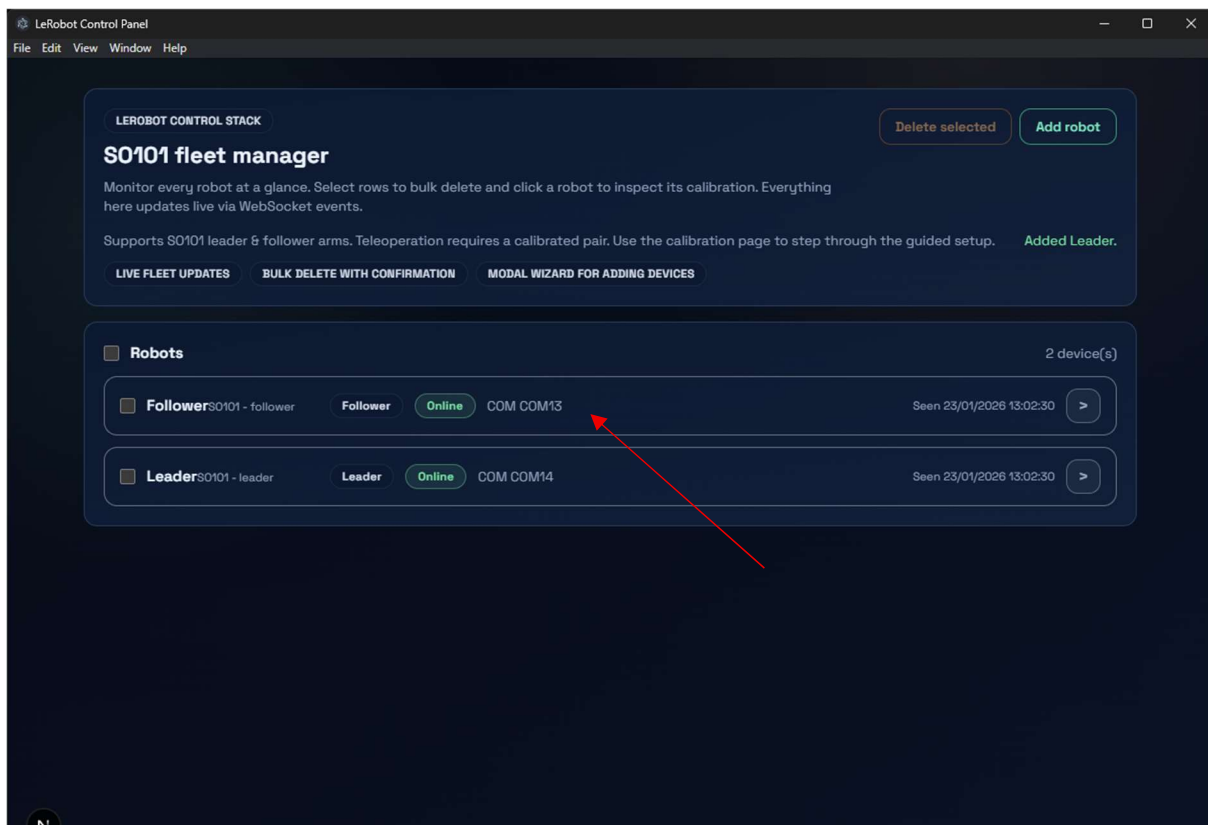
To do this, you need two calibrated arms, I've done this already for the "Follower" arm.

Do the same and follow these steps to control your robot arm.

Step 3.1: Access the specific robot page

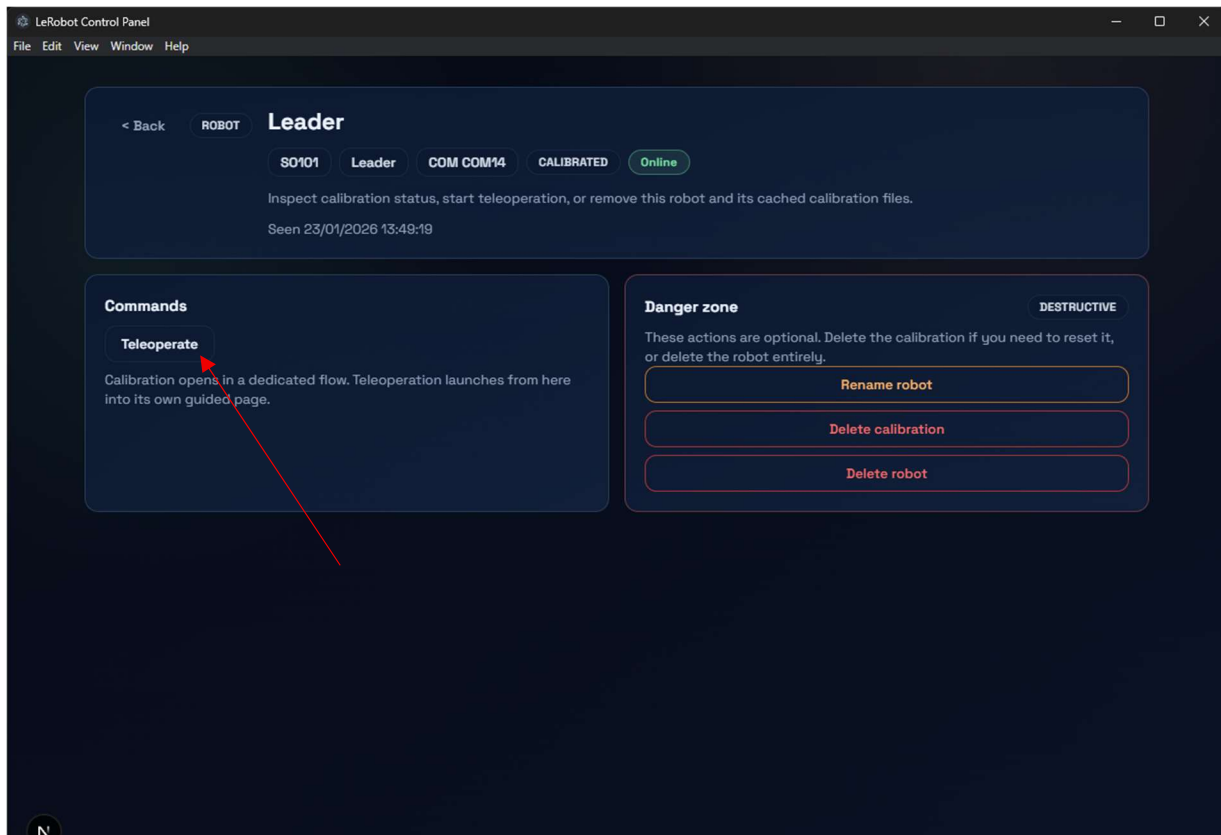
We need to access the specific page of the robot we want to control or control with.

Click on the item of the robot you want to control or control with.



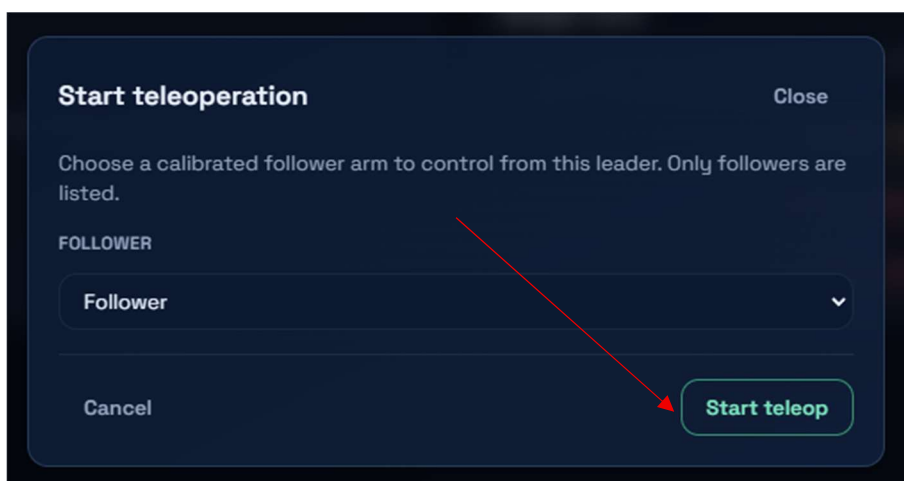
Step 3.2: Access teleoperation process

To access the calibration process you can just click on the “Teleoperate” button.



Step 3.3: Select the robot you want to control

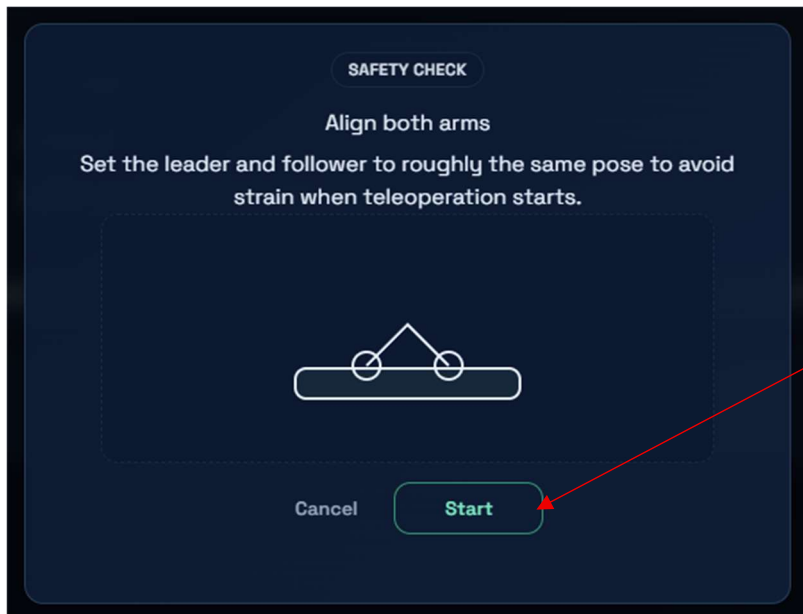
This page will open, here you need to select the follower, or leader arm. This option will be preselected but is useful when you have more than one follower or leader. Once you selected the arm you wish to control or the leader, you can press on the “Start teleop” button



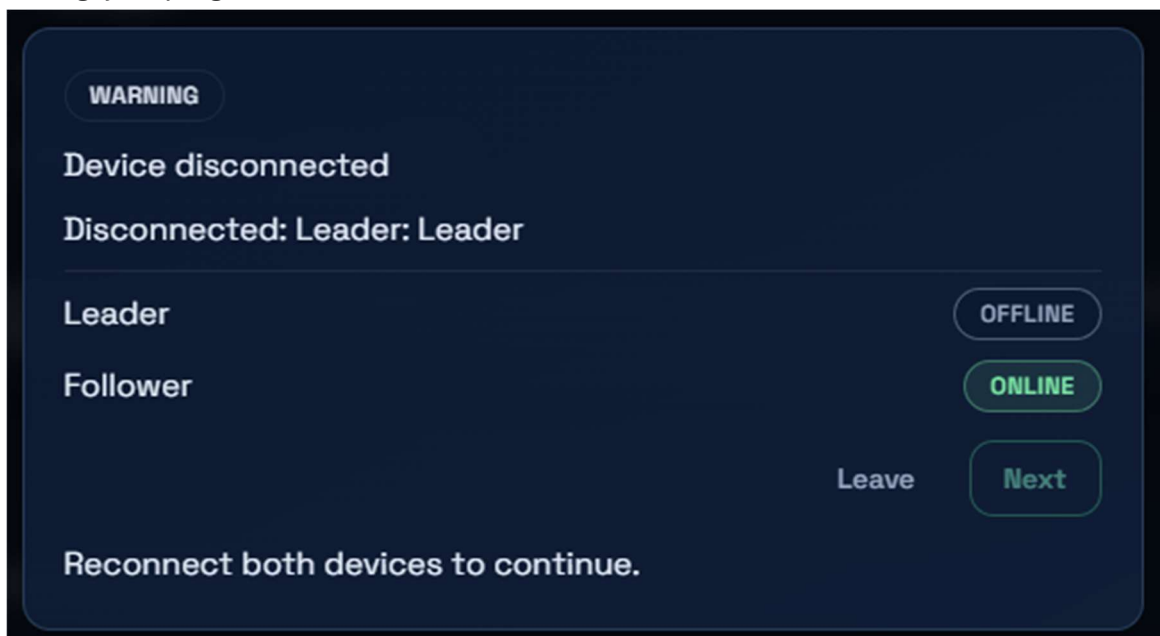
Step 3.4: Put both arms in same position

This dialog will ask you to place both robot arms in the same position. This is not required but strongly recommended. If not done the follower arm could swing to the same position as the leader arm and could add strain to motors, but mainly risk to kick over stuff on the table or damage cables.

Once this is done click on “Start” button

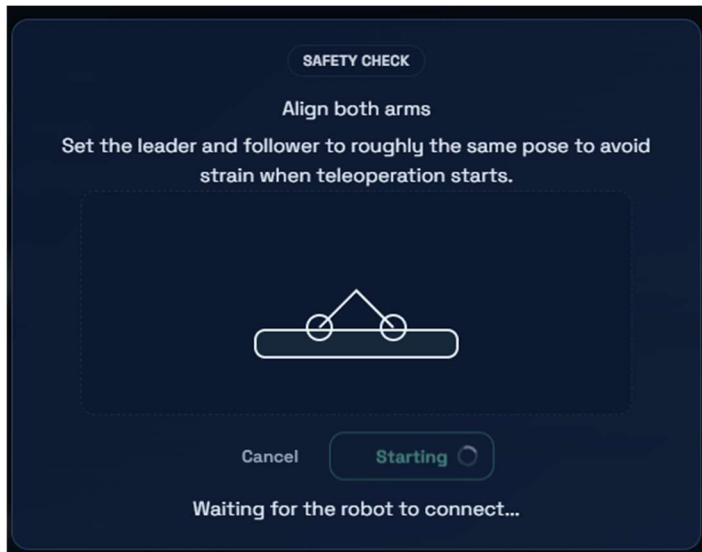


NOTE: if one of the arms is not plugged in and this offline, you will get the following dialog, just plug in the robot and click “Next”:

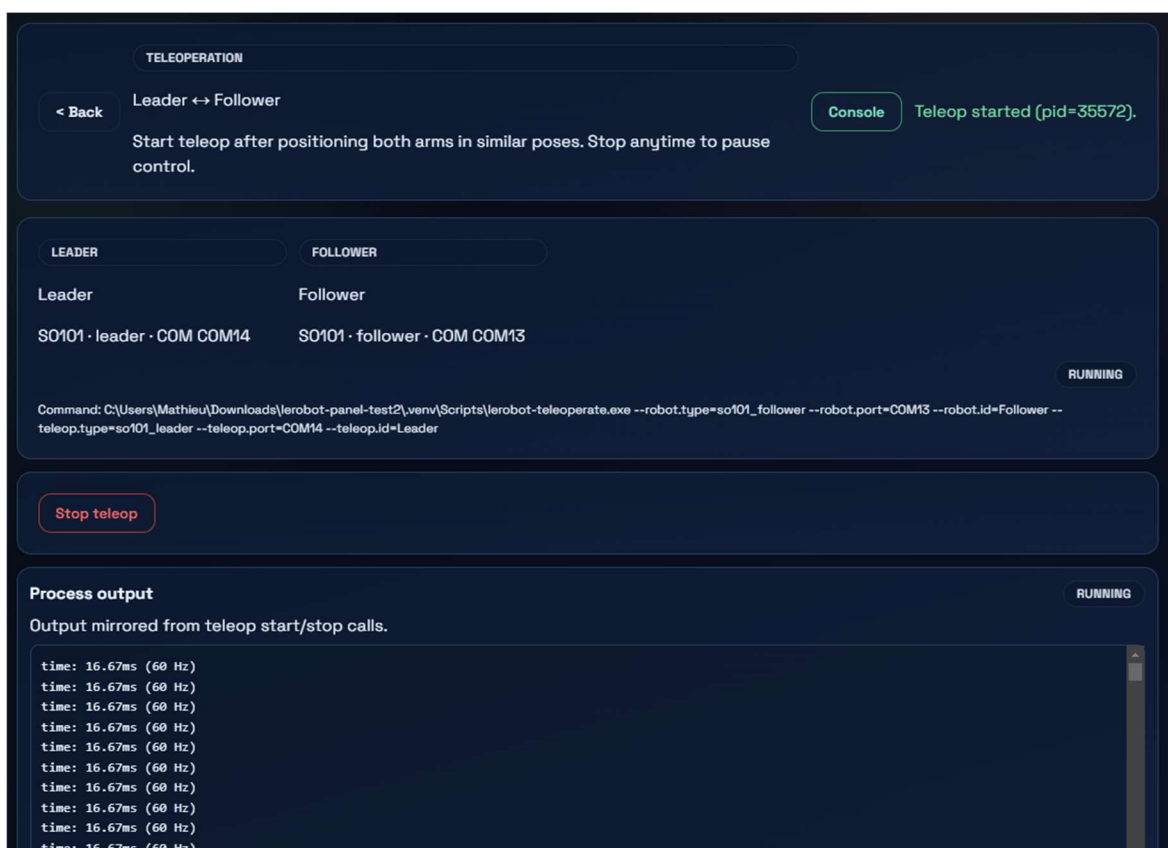


Step 3.5: Control the arm

Once the button is clicked the software will get ready, this might take a few seconds.

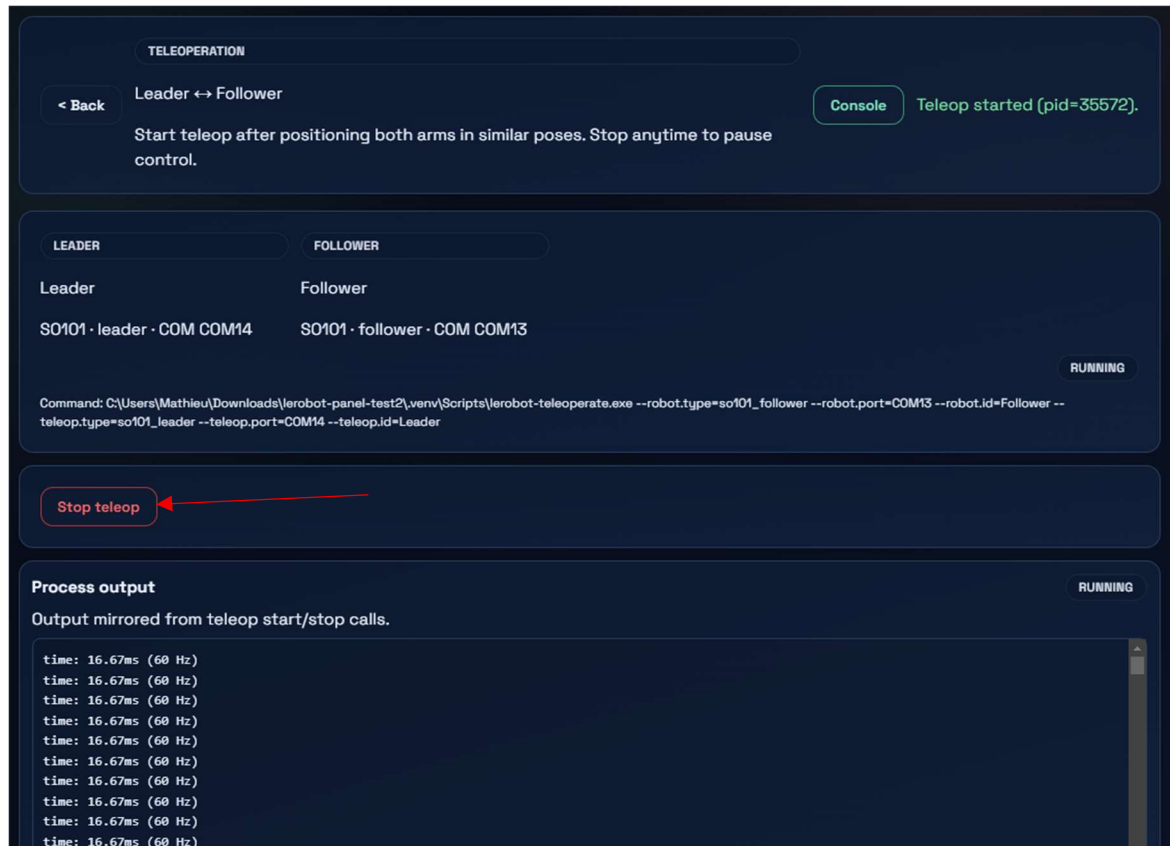


You should see this after some time; This confirms your controlling the arms, move the leader around to see the follower move.



Step 3.6: Stop teleoperation

Once you are satisfied and are done, you can simply click on the “Stop teleop” button. This will end the teleoperation.

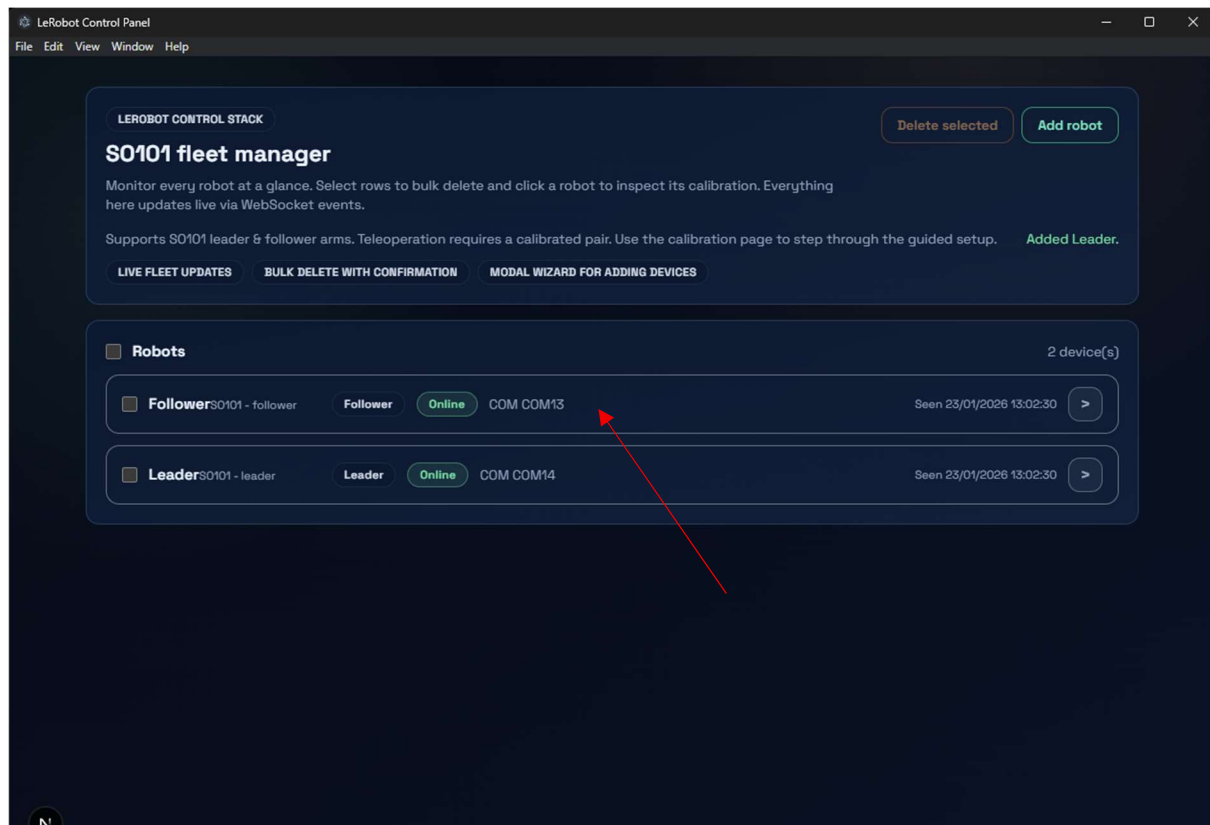


Step 4: Add cameras

Adding cameras is only possible on follower arms.

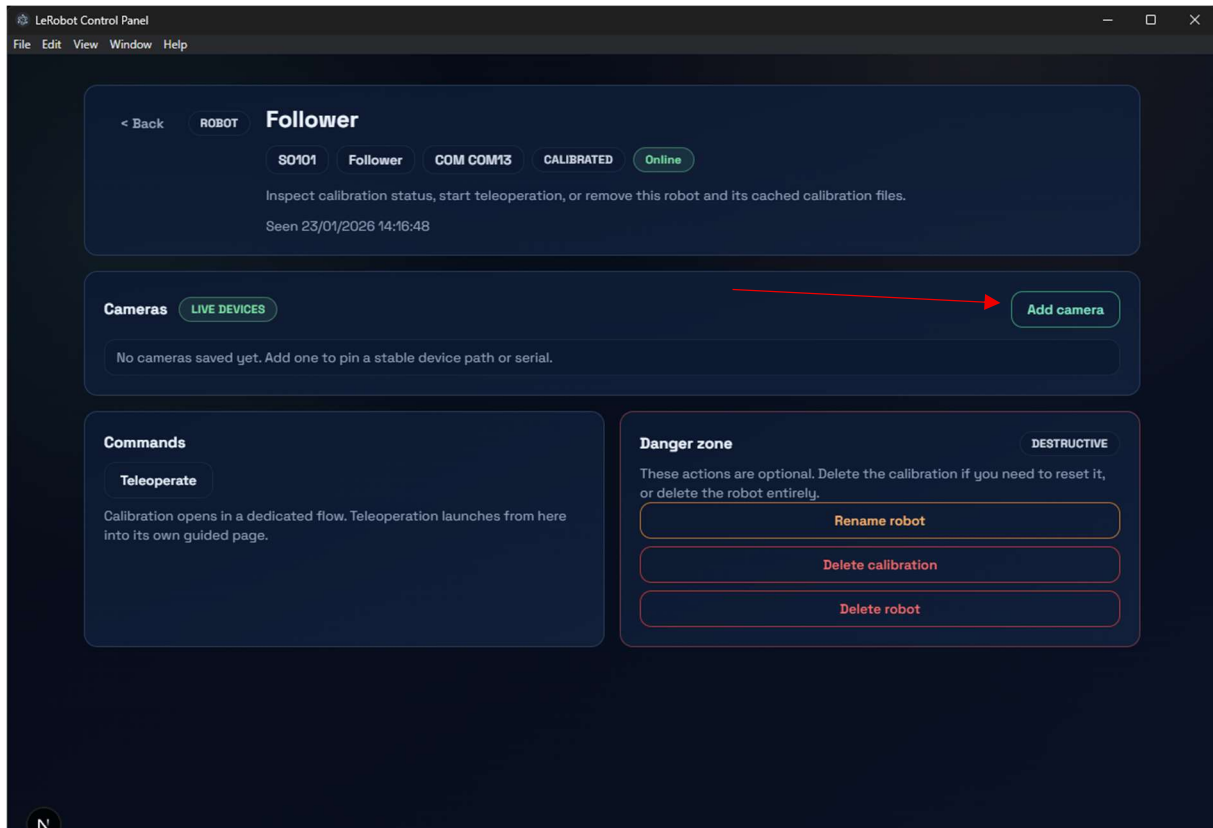
Step 4.1: Access the specific robot page

Compared to the teleoperation, here it does matter which robot we select. So, we need to select the robot arm with a type of “follower”



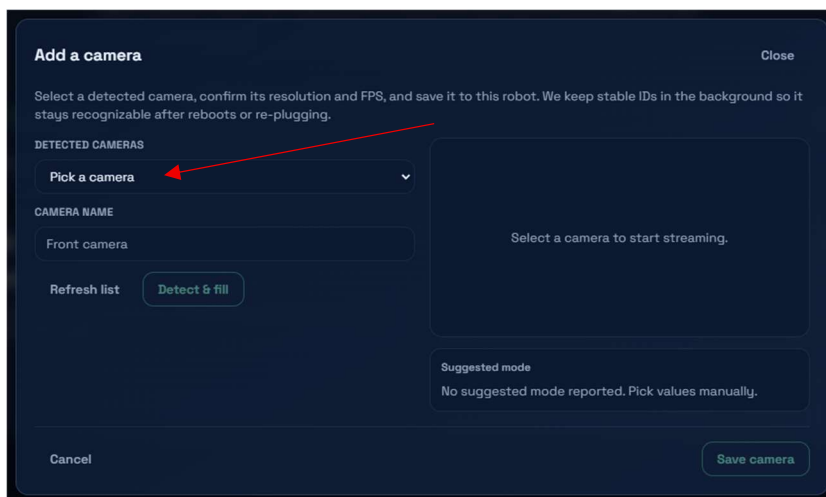
Step 4.2: Access the add camera process

To access the process please click on the “Add camera” button.

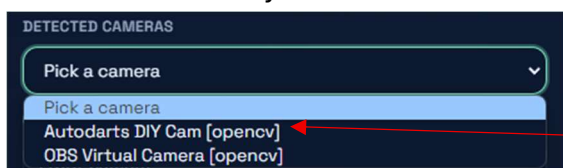


Step 4.3: Select the camera

There is a dropdown “Pick a camera”, select



This will then show you all detected cameras, select your webcam.



Step 4.4: Fill in a custom name

Once selected you will see the live feed of the camera, confirm this is the one you wish to select, and fill in a custom descriptive name. For example, here, “Grip” would be a good name

Add a camera Close

Select a detected camera, confirm its resolution and FPS, and save it to this robot. We keep stable IDs in the background so it stays recognizable after reboots or re-plugging.

DETECTED CAMERAS

Autodarts DIY Cam - global [opencv]

CAMERA NAME

Front camera

Refresh list Detect & fill

RESOLUTION

Width

Height

FPS

FPS

Restart stream Live MJPEG stream

Suggested mode

No suggested mode reported. Pick values manually.

Cancel Save camera

Step 4.5: Fill the camera specs

Next fill in the camera specs. This is advanced but is a required step, you can look this up on internet with reference to your webcam. If you're on sure and are on an ai computer there is another option under the following picture.

CAMERA NAME

Grip

Refresh list Detect & fill

RESOLUTION

Width

Height

FPS

FPS

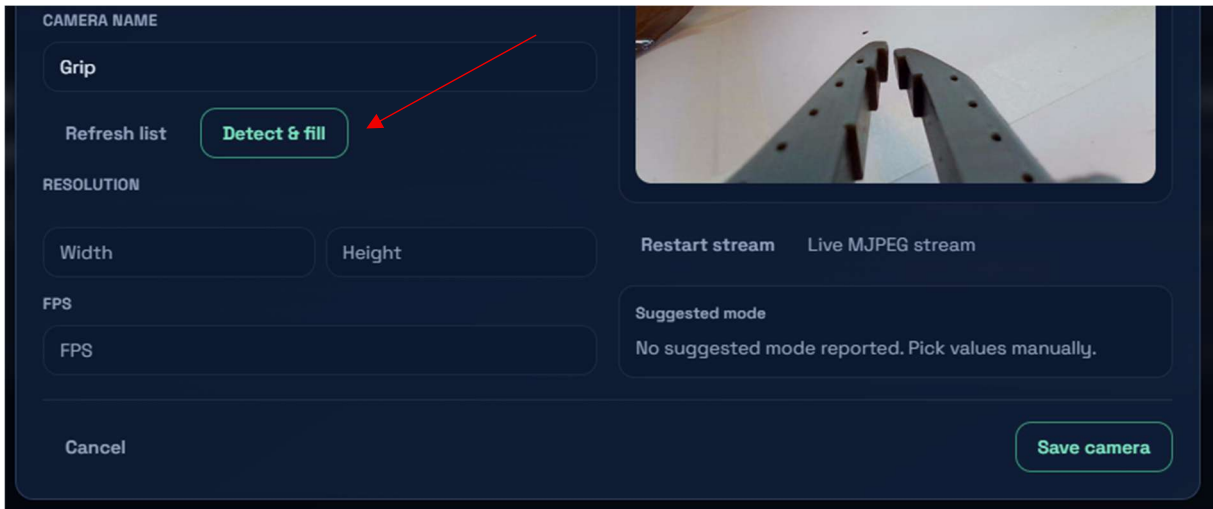
Restart stream Live MJPEG stream

Suggested mode

No suggested mode reported. Pick values manually.

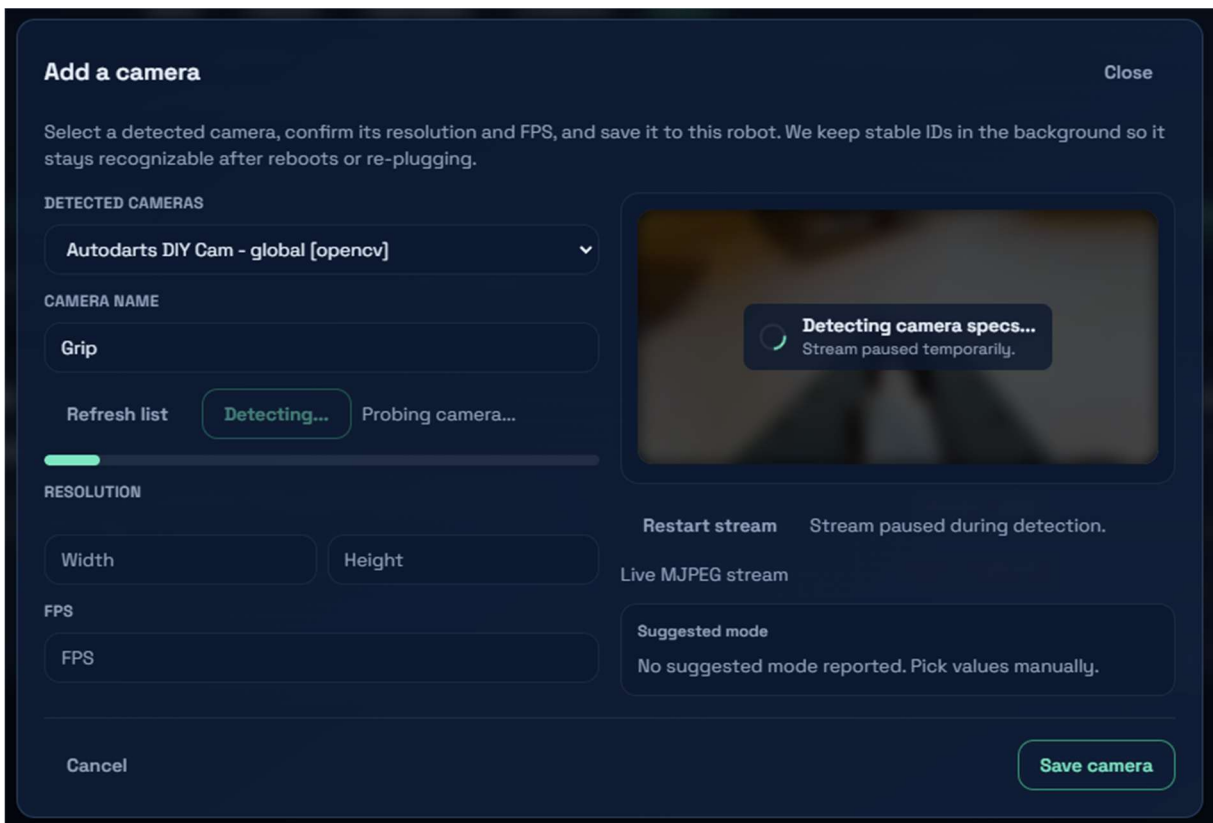
Cancel Save camera

You also have the option to click on the detect button, this requires a strong computer and might time out on any other machine.



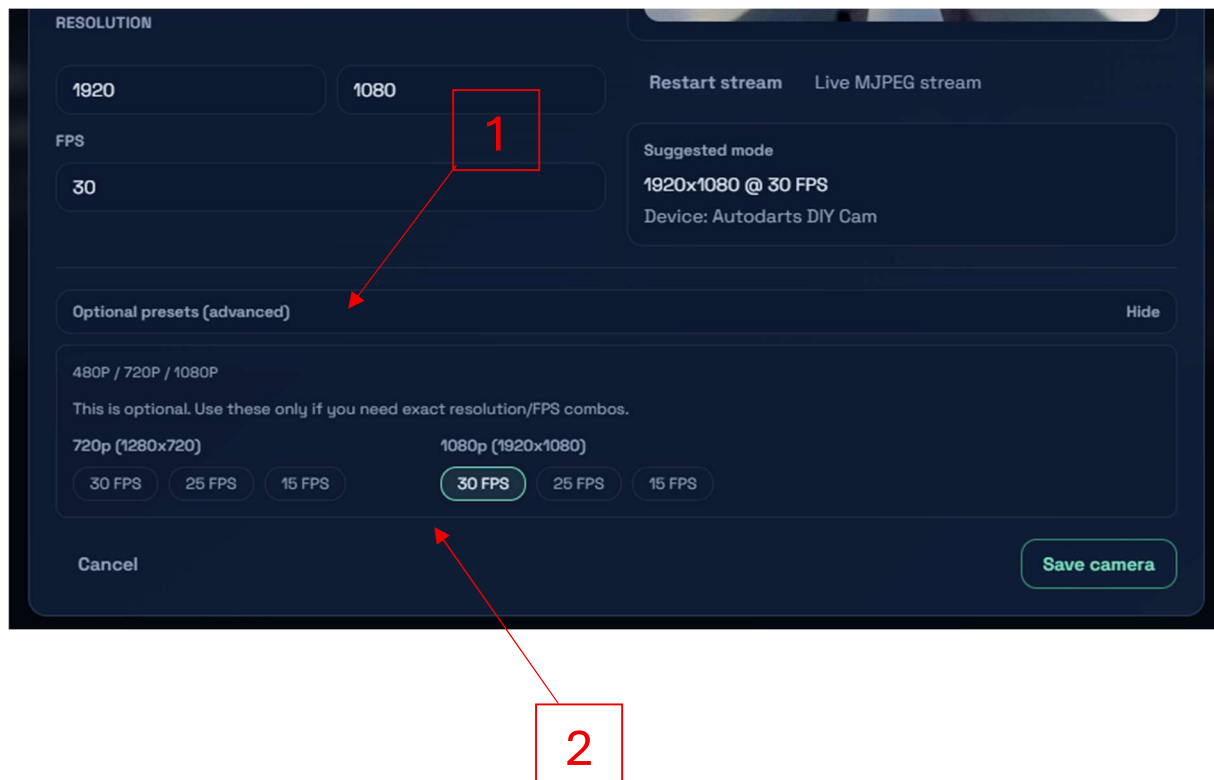
The screenshot shows a dark-themed dialog box titled "Add a camera". On the left, there are input fields for "CAMERA NAME" (containing "Grip"), "RESOLUTION" (with "Width" and "Height" sub-fields), and "FPS" (with an "FPS" sub-field). A "Refresh list" button is next to the camera name field. A red arrow points to a green "Detect & fill" button. On the right, there is a video feed showing a close-up of a mechanical part. Below the video feed are buttons for "Restart stream" and "Live MJPEG stream". At the bottom right is a green "Save camera" button, and at the bottom left is a "Cancel" button. A "Suggested mode" section at the bottom right states "No suggested mode reported. Pick values manually."

The process looks like this, and will detect all framerates for every resolution. This might take some time.



This screenshot shows the same "Add a camera" dialog box during the detection process. The "DETECTED CAMERAS" dropdown now shows "Autodarts DIY Cam - global [opencv]". The "CAMERA NAME" field still contains "Grip". The "Refresh list" button is now disabled, and the "Detecting..." button is active, with a progress bar below it. The text "Probing camera..." is visible next to the "Detecting..." button. The video feed on the right now displays a dark screen with a green circular progress indicator and the text "Detecting camera specs... Stream paused temporarily." Below the video feed, the "Restart stream" button is now disabled, and the "Live MJPEG stream" button is active. The "Suggested mode" section remains the same. The "Save camera" and "Cancel" buttons are still present at the bottom.

After a while you should get the following, the fields will get prefilled, but you can also go in the advanced tab and here you have other possible presets.



Step 4.6: Save camera

If everything is filled as you wish, you can proceed by clicking on the “Save camera” button. This then will add the camera to the robot. Good, this now means the camera will get used when starting teleopt.

